From Rental Endpoints to Road Traffic Counts: Assessing the Predictive Value of Car-Sharing Data

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ABSTRACT
Car-sharing records describe selective service activity rather than road traffic. We assess whether rental endpoints improve motor-vehicle count estimation at locations excluded from training. A Bologna case study combines 411,241 retained rentals with traffic counts from 2023–2024. Source audits reconcile detector identifiers and require complete hourly recording. Gradient boosting and regularised Poisson regression compare contextual inputs with citywide sharing activity, local endpoints, and origin–destination geometry. Five spatial folds group nearby devices and exclude training detectors within 250 m of test devices. Models fitted in 2023 are evaluated at excluded devices in both years. Citywide sharing reduces boosting's pooled weighted absolute percentage error from 62.9% to 62.1% in 2023; detailed spatial features provide no further gain. The advantage disappears in 2024. Near-zero detector totals destabilise mean device-level percentage errors. These findings demonstrate the need for source auditing, spatial validation, and subsequent-year testing before rental data support estimates of broader traffic flows.

KEYWORDS
Car-sharing; traffic volume estimation; inductive loops; selective mobility data; spatial validation; transferability

1. Introduction

Cities need observations of where and when motor vehicles move to monitor road networks and plan further measurements. Fixed detectors provide repeated counts at particular positions, but their coverage and reliability vary. Car-sharing records offer a different view: they identify where a rental begins and ends and how long the vehicle remains rented. These observations can describe a service in detail without representing traffic passing a particular road section.

The practical question is whether rental records help estimate motor-vehicle counts at locations whose counts are excluded from training. A large endpoint file does not answer this question by itself. Sharing users, service boundaries, parking arrangements, and vehicle availability can all affect the observed sample. Moreover, the interval between collection and return can include parking and intermediate stops. Expanding rentals to general traffic therefore requires more than multiplying their number by a market-share factor.

Car-sharing research has long recognised the relationship between demand and vehicle availability (Jorge & Correia, 2013). Empirical work also documents associations between use, location, and alternative transport options (Becker et al., 2017). Research on network-wide volume estimation from probe vehicles (Eisinga & Lorkowski, 2025) provides a related motivation for integrating selective mobility observations with fixed counts. However, rental endpoints are not vehicle trajectories. Their predictive value must be checked for the available observation type rather than inferred from results obtained with a richer source.

We evaluate that value through an audit-first comparison of contextual models and progressively more detailed sharing inputs. The study uses car rentals, municipal motor-vehicle counts, and weather from 2023–2024. The count source contains hundreds of devices, allowing grouped and buffered spatial validation rather than random splits of repeated observations. Models fit 2023 counts outside each test fold and predict at excluded devices in both 2023 and 2024. Sharing activity and weather are observed for the period being estimated: the task is spatial gap filling, not forecasting before those inputs become available.

The source audit is part of the evaluation, not a preliminary administrative step. The rental files differ between years, including the absence of recorded distance in

2024. The detector catalogue warns that changing codes can refer to the same physical device and supplies separate hourly completeness records (Comune di Bologna, 2026a, 2026b). Ignoring either issue would change the meaning of the features or target. Even a complete-recording flag does not independently verify a sensor's physical operation, a distinction that becomes important in the local-error diagnostics.

The contribution is a reproducible evaluation of selective rental features under spatial and subsequent-year transfer. Two model families use the same four cumulative input sets, and a separate scaling rule supplies an interpretable reference. We report pooled, device-level, and grouped errors because a model can fit busy detectors while failing at quieter ones. This is an empirical contribution in evaluation and interpretation, not a new prediction algorithm or a reconstruction of citywide travel demand.

Three questions organise the analysis. *RQ1*: which car-rental observations are usable, and what spatial and temporal patterns do they describe? *RQ2*: how much do citywide activity, local endpoints, and endpoint geometry add to a contextual count model? *RQ3*: do the resulting estimates transfer to excluded spatial locations and a subsequent year? Sections 2.1 and 3.1 address RQ1 through the source audit and descriptive results. Section 2.2 describes the predictive methods, and Section 3.2 addresses RQ2 through matched feature comparisons and RQ3 through spatial and subsequent-year tests. Section 4 discusses the observation and interpretation limits, and Section 5 answers the three questions.

1.1. Related Work

1.1.1. Car-Sharing Demand and Recorded Rentals

Jorge and Correia (2013) review car-sharing demand estimation and operations, emphasising that observed use and vehicle supply are linked. A rental dataset therefore records realised service activity under its operating conditions, not unconstrained demand for all motorised travel. Becker et al. (2017) analyse free-floating car-sharing transactions in Switzerland using spatial regression and conditional logit models. Their study supports examining the spatial context of service use; it does not make rentals a probability sample of all traffic.

The distinction matters for endpoint summaries. A return close to the collection point can represent a genuine round-trip rental. A long rental can contain periods when the vehicle is stationary. Arrivals minus departures describe recorded rental flows, but do not determine available vehicles without opening stocks and operator movements. We therefore retain near-origin rentals and distinguish rental duration from driving time. The descriptive analysis establishes what the inputs represent before asking whether they help predict detector counts.

Bicocchi et al. (2017) analyse urban mobility traces to identify matching rides along similar routes and support ride-sharing recommendations. Their study illustrates the use of recorded movements to inform shared-mobility services, but ride-sharing opportunities are a different target from car-rental activity or total road traffic. Here, we assess rental features against independent motor-vehicle counts rather than infer their usefulness from service-level applications.

1.1.2. Selective Vehicle Data and Traffic Volumes

Network-wide traffic volume estimation from probe vehicles is an active research problem. Eisinga and Lorkowski (2025) study volume estimation based on probe vehicle

data, illustrating the broader objective of using selective mobile observations beyond fixed counting positions. The observation mechanism remains essential: trajectories can provide evidence of link use, whereas our rental records expose only endpoints and times. A straight line between those endpoints is a geometric predictor, not an observed road crossing.

For the same reason, our comparison is not framed as a general test of probe-data traffic estimation. It evaluates particular representations of a particular rental source. A contextual baseline is necessary because calendar, weather, and position can already explain part of the target variation. A matched comparison within each model family then measures the added value of sharing rather than differences in training samples or model complexity alone. The contextual baseline is not claimed to exhaust all available road information: validated lane counts, road classes, and access constraints are not supplied as predictors here.

Spatial validation must also reflect the intended use. Splitting individual dates at random would leave observations from the same detectors in both training and testing. Nearby detectors could create a similar problem even when device identifiers differ. We instead group spatially connected devices, exclude whole kilometre blocks, and remove nearby training devices through a buffer. This tests estimation away from training sensors rather than interpolation over missing times at known devices.

1.1.3. Count Prediction and OD Inference

Inferring an origin–destination (OD) matrix from traffic counts has a long methodological history. Entropy-based estimation (van Zuylen & Willumsen, 1980) and generalised least squares (Cascetta, 1984) combine prior information with count constraints. More recent Bayesian work explicitly incorporates traffic assignment and route probabilities (Hazelton & Najim, 2024). These methods address a different target from the direct count models used here.

Mamei et al. (2019) examine OD estimation from mobile phone call detail records, including time-based and routine-based approaches, population scaling, and projection onto the road network. Their work provides a related evaluation perspective on deriving broader movement estimates from source-specific observations. Our rental endpoints enter direct count models instead, without reconstructing population-wide demand or assigning trips to road paths.

A count fit alone does not establish that an estimated OD explanation is correct. Different movements and route choices can produce similar observations at a limited set of sensors. We therefore use rental OD geometry as an input rather than reconstructing a total-demand matrix. Predictions are checked against measured detector passages; no reference trajectories or independent OD observations are available to validate the journeys that generated them.

2. Materials and Methods

2.1. Car-Sharing Data and Rental Patterns

The experimental setting is Bologna, Italy. Two supplied files contain car rentals beginning in 2023 and 2024. This section describes the source audit and the methods used to characterise rental timing, endpoint distributions, and arrival–departure balances. The descriptive results in Section 3.1 characterise the observed service, not all motor-vehicle travel.

2.1.1. Data and Filtering

The source files contain 453,945 rows. Both include a user code, number plate, start and return timestamps, and endpoint coordinates. Recorded distance is present only in 2023. We use identifiers only for exact duplicate checks and remove them before writing the trip cache. Original source files remain unchanged. Precise endpoints remain potentially sensitive even after identifiers are removed and are not included in released numerical results.

We parse the local timestamps and decimal-comma coordinates, then retain records with a departure in the source year, both endpoints inside latitude 44.40–44.60 and longitude 11.20–11.50, and a finite recorded duration of at least 30 seconds. Boundary values are included. This box defines the analytical area rather than the municipal boundary. Rentals outside it are excluded from this analysis, not classified as erroneous. The same rules apply in both years.

The filtering procedure retains 411,241 rentals, about 195,000 in 2023 and 216,000 in 2024 (Table 1). No exact duplicates or records failing the duration criterion are found under these rules; the reduction is due to the endpoint box. A minimum endpoint-displacement rule would remove plausible round trips, so it is not applied. About 4% of retained rentals end within 100 m of their starting position. Neither a near-origin return nor a long rental is by itself evidence of an invalid observation.

Endpoint displacement uses a local equirectangular projection centred at $(44.4949, 11.3426)$, with distances expressed in kilometres. It is a straight-line separation, not the distance driven. Recorded rental duration is return time minus start time and may contain stops. The main analysis does not cap it. Supplementary summaries use upper limits of two, six, and 24 hours; the prediction sensitivity experiment rebuilds every sharing feature using only rentals lasting at most six hours.

2.1.2. Monthly and Hourly Use

We aggregate retained departures by calendar month and compare departure-hour profiles by year and day type.

For both the sharing/count comparison and day/night descriptions, day covers 06:00–22:59. Night comprises 00:00–05:59 and 23:00–23:59 on the *same calendar date*. It therefore combines early-morning and late-evening hourly bins, rather than forming a continuous overnight service period. This convention matches the daily detector exports without assigning early-morning counts to another source date. Figure 2 separates weekday and weekend profiles, with each year/day-type group normalised to sum to one.

2.1.3. Displacement, Duration, and Near-Origin Returns

We summarise endpoint displacement, rental duration, and the shares of near-origin returns and rentals lasting more than 24 hours by year and day/night period (Table 2). Since 2024 has no recorded-distance field, we do not compare reported kilometres across years or impute them from duration. Even in 2023, reported distance is not independently verified and does not reveal a route.

2.1.4. Origin–Destination Patterns

We assign endpoints to square cells and count rentals for each origin–destination pair, including within-cell movements. To compare day and night independently of their

different trip totals, each matrix is normalised by its total. Jensen–Shannon distance with base-two logarithms is

$$\begin{aligned} \mathrm{JSD}(p,q) &= \sqrt{\tfrac{1}{2}D_{\mathrm{KL}}(p\|m) + \tfrac{1}{2}D_{\mathrm{KL}}(q\|m)}, \\ m &= \tfrac{1}{2}(p+q). \end{aligned} \tag{1}$$

It ranges from zero for identical distributions to one for disjoint supports.

Sparse OD tables can produce nonzero distances through sampling alone. Supplementary diagnostics therefore draw 20,000 observations from each period's empirical distribution, thirty times, and compare them with equally sized draws from their pooled distribution. These are descriptive sampling checks, not confidence intervals for population OD demand. Neither the raw distance nor the check identifies an expansion factor from sharing to all traffic.

2.1.5. Activity and Arrival–Departure Balances

Figures 4 and 5 pool 2023–2024 endpoint events in 500 m cells. They use four windows: 06–10, 10–15, 15–20, and 20–06. For consistency with calendar-date aggregation, the last combines 00–06 and 20–24 on a date. The windows are descriptive choices, not optimised thresholds. Departures use collection time and arrivals use return time, so a rental can contribute to different windows or dates at its two ends.

The fixed 24-bin detector exports do not distinguish repeated and missing clock hours. We exclude the four Europe/Rome daylight-saving transition dates from maps and count experiments rather than infer the meaning of their ambiguous bins. The remaining dates have ordinary 24-hour exposure. Events outside the two study years are not counted as arrivals within them. Each map panel uses its window length multiplied by the number of included dates, and cells below fifty endpoint events are suppressed in both map types.

For cell z and window r, activity is $(A_{zr} + D_{zr})/H_r$, where A and D denote arrivals and departures and H_r is the included exposure. The normalised balance is

$$b_{zr} = \frac{A_{zr} - D_{zr}}{A_{zr} + D_{zr}}. \tag{2}$$

Positive values indicate more returns than collections; negative values indicate the reverse. Activity and balance answer different questions. A busy cell may be balanced, while a less active cell can have a large relative imbalance.

2.2. Vehicle Count Estimation

We now test whether the audited rental information helps estimate independently recorded traffic. The target is the number of motor-vehicle detections by one device during a complete day or night period. The source does not classify passenger cars separately or identify sharing vehicles. We therefore use *motor-vehicle counts*, not car-only counts, as the target throughout the evaluation.

2.2.1. Detector Identities and Measurement Completeness

The municipal files contain 313,067 device-day rows for 2023 and 309,561 for 2024, with 24 hourly count fields and detector metadata. The catalogue explicitly notes

that a detector's code can change and identifies `ID_univoco_stazione_spira` as the stable station identifier (Comune di Bologna, 2026b). We use that identity rather than treating every code or level as an independent location. Levels in this source describe changing identifiers, not road hierarchy.

We retain active loop detectors with valid identities and coordinates and finite non-negative hourly counts. Conflicting count vectors for the same device/date are discarded. Identical repeated vectors under aliases are counted once, not summed. The supplied snapshots contain 999 such repeated rows in 2023 and 674 in 2024 after the preceding checks. Four station identities whose valid positions differ by more than 100 m from their pooled median position are excluded from the prediction cohort; this prevents a device with an unstable recorded position from serving as a fixed spatial target.

The count catalogue links separate accuracy files (Comune di Bologna, 2026a). Their documentation defines 100% as measurement throughout the reference hour, zero as no measurement, and intermediate values as partial coverage. We cache both official annual exports and join them by source date and loop code before resolving aliases. Every constituent hour must have a value of exactly 100% for its period to be usable. Partial periods are not scaled up, and missing or incomplete recording is not interpreted as zero traffic. If identical aliases provide different completeness, the record with the most fully recorded hours is retained, using the loop code to break ties deterministically.

This restriction removes many periods (Table 4) and changes the observation population. Completeness is the municipality's recording-status field, not an independent validation of every detection. Residual zero and near-zero series are retained rather than removed after inspecting model errors. Their influence is explicitly considered below.

2.2.2. Eligible Devices and Spatial Splits

Devices must have at least 180 distinct dates with a usable period in 2023. This coverage criterion is fixed using the training year, not 2024 outcomes. It leaves 752 eligible devices. Table 5 reports the usable periods at these devices in each year. The 2024 test uses available observations from the same eligible identities and does not require every device to remain observed.

Devices within 100 m are linked into connected components, giving 246 spatial groups. Each group's centroid is assigned to a fixed 1 km grid cell, yielding 63 occupied blocks. Coordinates x, y are in kilometres from the reference point. A block with integer indices (g_x, g_y) belongs to fold $(g_x + 2g_y) \bmod 5$. This deterministic allocation depends on position, not observed volume or prediction error. All devices in a connected component and all components in a block share a fold.

For each test fold, we train on eligible devices in the other four folds, additionally excluding any training device within 250 m of a test device. Training uses 2023 only. The fitted model predicts the held-out fold in 2023 and, without refitting, its available 2024 observations. The five training sets contain 494–636 devices. Neither local count histories nor neighbouring excluded counts enter the predictors. Figure 6 shows the allocation; the pipeline saves device lists and verifies the minimum train–test separation.

The 2023 test evaluates buffered spatial transfer within the observed year. The 2024 test combines spatial exclusion with subsequent-year change. It does not isolate time transfer at previously observed devices, and differences in valid-period coverage between years can contribute to its aggregate results.

2.2.3. Models and Cumulative Feature Sets

Models predict a mean count per hour for each device/date/period, then multiply by the seventeen daytime or seven nighttime hours. Exposure hours are also training weights. Four cumulative input sets are used within both gradient boosting and regularised Poisson regression:

(1) *No sharing*: projected coordinates, distance from the reference point, weekday, annual sine and cosine terms, day/night indicator, holiday indicator, daily mean temperature, and precipitation.
(2) *Citywide sharing*: the first set plus retained rental departures per hour across the analytical area.
(3) *Local endpoints*: the second set plus collections and returns within 500 m of the detector, per hour.
(4) *OD geometry*: the third set plus departures whose endpoint-to-endpoint chord passes within 100 or 250 m of the detector, per hour.

Holidays include Italian public holidays and Bologna's patronal day. Weather is joined by calendar date. All feature sets use identical count observations. Rental features use the entire retained cohort, not a subset selected according to prediction success.

For endpoints o, d and sensor position s, chord proximity is defined by

$$\begin{aligned} u &= \frac{(s-o)^\top (d-o)}{\|d-o\|^2}, \\ \delta &= \|o + \min(1, \max(0, u))(d-o) - s\|. \end{aligned} \tag{3}$$

A rental contributes only if $0 < u < 1$ and δ is within the specified radius. A zero-length chord does not contribute. These unsigned features do not assign a verified driving direction or road crossing. Returns use their own date and hour for endpoint features; chords use the departure period, even when the rental continues later. This deliberately simple representation cannot recover the timing of intermediate movement.

Boosting uses Poisson loss, 120 iterations, seven leaves per tree, learning rate 0.06, minimum leaf size forty, and L_2 penalty ten, without early stopping. Poisson regression is log-linear with L_2 penalty one, at most 1,000 optimisation iterations and tolerance 10^{-7}. Weekday and period are one-hot encoded for Poisson; numeric inputs are median-imputed and standardised using training data only. The same preprocessing is applied unchanged to test observations. Settings are fixed across feature sets and are not selected from test counts.

A separate *uniform expansion* baseline multiplies the 250 m chord rate by a non-negative factor that matches the exposure-weighted total training count. It has no contextual adjustment and predicts zero when the chord feature is zero. The primary comparison therefore has nine configurations: two families with four input sets each, plus scaling. Two further fits, one per family, rebuild the complete sharing input set using rentals no longer than six hours. They test sensitivity to long rental intervals on the same count samples, rather than deleting test observations.

2.2.4. Error Measures and Uncertainty

For device s with a positive observed total,

$$\mathrm{WAPE}_s = \frac{\sum_{i \in s} |y_i - \widehat{y}_i|}{\sum_{i \in s} y_i}. \tag{4}$$

We report *macro* WAPE, the mean over devices with positive denominators, and *pooled* WAPE, the sum of all absolute errors divided by the total number of observed detections. All-zero devices have undefined device WAPE and are omitted only from macro and median percentage summaries; their absolute errors remain in the pooled calculation. No denominator is replaced with an arbitrarily small constant. We also report the median device WAPE and *cluster* WAPE, which pools errors and detections within each 100 m connected component before averaging across components with positive totals.

These measures have different meanings. Pooled error gives more weight to busy detectors and counts repeated passages at different devices separately; it is not an error relative to unique citywide journeys. Cluster averaging reduces the multiplicity of nearby devices but can hide differences between them. Macro error treats devices equally but becomes unstable when some denominators are close to zero. Signed bias, $(\sum \widehat{y} - \sum y)/\sum y$, distinguishes systematic over- and underestimation from unsigned error.

Paired feature contrasts compare each sharing model with the no-sharing model of the same family. We resample the occupied 1 km test blocks with replacement 2,000 times, retaining their device-level WAPE differences, and report percentile intervals for the mean difference. This avoids treating repeated periods as independent spatial samples. It does not eliminate dependence across adjacent blocks or overlapping training sets; intervals are descriptive measures of spatial variation, not simultaneous hypothesis tests or citywide uncertainty bounds.

3. Results

3.1. Rental Patterns

Figure 1 shows monthly retained departures. The profiles differ between years: rentals are more numerous early in 2023, whereas 2024 has a marked autumn increase. These observations cannot distinguish changes in customers' travel from changes in fleet availability, service rules, or the supplied data coverage. Unlike an independently documented service census, a rental file does not establish that every potential journey could be observed.

Night departures account for about 16% of retained rentals in each year. The hourly profiles in Fig. 2 identify when this service is used, but not the purpose of individual rentals. An evening peak should not automatically be interpreted as commuting, and weekend departures need not represent a single type of leisure activity. Such explanations would require trip-purpose or user information beyond the endpoint records used here.

Table 2 reports quantities available in both years. Median endpoint displacement is about 2.5–2.8 km across the four year/period groups, while median rental duration ranges from about sixteen to 25 minutes. The 2024 medians are longer than the corresponding 2023 medians for both measures. This is a comparison of retained records

under a common spatial filter, not a claim of longer routes or changed driving speed.

Figure 3 shows the distributional tails behind these medians. Long rentals are uncommon but cannot be treated as uninterrupted driving. More than 24 hours separate collection and return for about 1–2% of rentals within the year/period groups.

At 1 km resolution, the Jensen–Shannon distance between day and night OD patterns is about 0.34 in 2023 and 0.33 in 2024 (Table 3). Coarser cells reduce the difference. In the sampling check at 1 km, the mean distances are about 0.43 for separate-period draws versus 0.31–0.32 for pooled-distribution draws. Thus the observed period contrast is not reproduced by this common-distribution sampling check, although sparsity contributes substantially to the measured distance.

The maps in Figs. 4 and 5 describe endpoint concentrations and net rental movements. Without inventories, vehicle status, and operator relocation logs, they do not establish how many rentable cars remain in a cell. Nor can an endpoint hotspot be read as a traffic hotspot: a rental starting nearby may use another street, and traffic through a detector may have neither endpoint nearby.

3.2. Prediction Results

Tables 6 and 7 report the primary comparisons. For boosting, citywide sharing reduces pooled 2023 WAPE from 62.9% to 62.1%, a reduction of less than one percentage point. Under the fixed configuration, adding local endpoints or chord features produces exactly the same predictions as citywide sharing alone. The richer feature sets are therefore available to the learner but do not change its fitted predictions in this experiment.

Poisson regression has pooled 2023 WAPE of 64.9% without sharing, 64.6% with citywide activity, 64.6% with endpoints, and 64.6% with geometry. These changes are small relative to the remaining error. Uniform expansion is worse on pooled and cluster summaries, showing that a single chord-to-count factor cannot account for the observed local differences.

In 2024, boosting's pooled WAPE is 61.6% without sharing and 61.7% with any of the primary sharing sets. Poisson's error rises from 63.6% without sharing to 68.8% with citywide sharing and 69.0% with geometry. The modest 2023 pooled gain therefore does not establish a transferable sharing advantage. Nor does the larger volume of 2024 rentals imply better count predictions.

The large macro values require a different interpretation from the pooled results. Some detectors have almost no observed traffic despite many officially complete periods. For example, eligible device 103 records only one detection across 501 valid periods in 2023. A nonzero prediction there yields an enormous percentage error. This observation does not prove whether the device is malfunctioning, measuring a restricted movement, or described incompletely by the metadata. It does show that recording completeness alone cannot make mean percentage error a stable summary.

Table 8 and Fig. 7 therefore accompany the uncapped macro results. Cluster errors remain high even after nearby devices are pooled, so unstable denominators are not the only problem. Conversely, the pooled signed biases can be small while absolute errors are large: overprediction at some positions cancels underprediction at others in the total, but not in WAPE.

The block contrasts in Table 9 expose this denominator sensitivity rather than resolving it. Very large 2023 mean changes can coexist with a small pooled change. The 2024 Poisson sharing contrasts are unfavourable, consistent with their higher pooled

and cluster errors, but their magnitudes depend strongly on low-volume devices. These results answer RQ2 conditionally: sharing produces only small pooled improvements in the within-year test, and additional endpoint detail is not reliably useful. For RQ3, the improvements do not persist in the subsequent-year test.

3.3. Sensitivity and Local Calibration

Table 10 reports models refitted with the complete feature set derived from rentals lasting at most six hours. This removes about 5–7% of the main rental cohort while keeping all count test rows. Pooled errors remain close to the corresponding full-cohort geometry results. Long rental intervals therefore do not, by themselves, explain the broad lack of a transferable advantage. This tests sensitivity to one representation choice; it does not prove that all rental cleaning choices are unimportant.

Figure 8 separates changes over time from local volume calibration. Four devices are selected using the 12.5th, 37.5th, 62.5th and 87.5th percentiles of observed 2023 mean flow, not according to model error. Weekly rates use the same available test periods for observations and predictions, without filling unavailable weeks with zeros. The profiles illustrate that plausible seasonal variation can coexist with persistent errors in the level of a particular device.

The examples illustrate distinct calibration problems. At device 366, boosting with geometry overestimates the 2024 total by about 167%, while Poisson regression with the same inputs overestimates it by about 439%. At device 801, boosting with geometry instead underestimates the total by about 50%. Device 722 has a positive total bias of only about 2% with boosting and geometry, but still has WAPE of about 16%: cancellation in the total does not imply accurate period-level estimates. The Poisson geometry profiles also show autumn peaks that are much stronger than the corresponding measured profiles at several devices. Their timing is compatible with a response to the observed sharing increase, not proof of a real increase in local traffic.

The common issue is the difference between activity near a sensor and traffic detected by it. Nearby endpoints can involve parallel streets, while through traffic can originate and terminate elsewhere. Unsigned chords ignore road access, one-way restrictions, detours, and intermediate stops. Spatial coordinates alone also omit lane and approach characteristics. These are plausible reasons for imperfect transfer, not identified causes of any particular residual. Verified routes and detector-to-road mappings would be needed to test them.

4. Discussion and Limitations

The results do not support treating the observed rentals as a fixed fraction of all motor-vehicle passages. The scaling baseline performs poorly, and the learned models gain little from richer endpoint detail under the tested representations. This is not evidence that car sharing is generally unrelated to traffic. It is evidence that these features, models, and validation conditions do not establish a stable improvement over contextual inputs.

The audit reveals two distinct uncertainty sources. Rental records selectively observe a service, and the endpoint box deliberately omits journeys crossing its boundary. Detector records selectively observe instrumented positions and periods, with substantial losses under the complete-recording requirement. Neither source has a known sampling probability for all urban movements. Their combination cannot remove those selection

mechanisms automatically.

The quality flags make incomplete recording visible, but are not independent ground truth. Residual near-zero series dominate unweighted percentage averages, and all-zero devices require an explicit denominator convention. We report these cases rather than selecting a traffic threshold after observing prediction errors. Before operational deployment, provider-supported validation should establish whether these devices describe genuine low-volume movements or instrument problems. Alternative observation-error models and absolute-error summaries would then be appropriate extensions.

The spatial design is stronger than a random split of repeated dates, but its units are still within one city and one sensor system. Connected components can contain several devices and do not necessarily correspond to a single road section. A 250 m buffer prevents very close training observations, not all network dependence. The same spatial folds are used for all comparisons, while their overlapping training sets limit formal independence. More cities and independent sensor networks are needed for external validation.

The comparison also concerns fixed model configurations, not the best achievable estimator. More extensive spatially nested tuning, validated road attributes, and other traffic observations could strengthen both sharing and non-sharing models. Identical boosting predictions after local inputs are added mean that this learner did not exploit them under the chosen settings; they do not prove that those variables contain no information under every model. The 2024 decline in Poisson sharing performance likewise cannot be attributed uniquely to a change in travel behaviour, service coverage, or road policy.

No causal effect of car-sharing use on total traffic is estimated. Rentals and traffic can respond jointly to calendar conditions, service supply, and unobserved events. Nor do detector predictions identify unique journeys or an OD matrix. A defensible citywide flow map would require validated road-section targets and broader spatial evidence; a citywide demand matrix would additionally require route assignment and independent movement validation.

5. Conclusion

This study evaluated car-sharing features for motor-vehicle count estimation through source auditing, matched feature comparisons, and buffered spatial and subsequent-year tests.

RQ1—Usable records and rental patterns. Common filters retain 411,241 rentals from the two supplied files. Near-origin returns are preserved, rental duration is distinguished from driving time, and distance comparisons across years use endpoint displacement because recorded kilometres are absent in 2024. Temporal profiles, OD comparisons, and activity/balance maps describe the sharing sample without assuming it represents all traffic.

RQ2—Incremental predictive value. Citywide sharing reduces boosting's pooled 2023 spatial-test WAPE from 62.9% to 62.1%, while local endpoints and geometry do not alter its predictions under the fixed configuration. Poisson's pooled improvements are also small. Uniform chord expansion performs worse. More detailed rental features are therefore not, by themselves, evidence of substantially better traffic estimates.

RQ3—Transfer beyond training observations. The modest 2023 pooled benefit does not persist in 2024: boosting has WAPE of 61.6% without sharing and 61.7% with it, while sharing worsens the Poisson results. Residual near-zero and zero detector series

further show why complete recording, stable percentage summaries, and correct local calibration are different requirements.

The broader methodological lesson is to evaluate selective endpoint data against contextual baselines, reconcile changing sensor identities, use measurement-completeness information, exclude nearby training positions, and inspect multiple local and aggregate errors. The Bologna use case demonstrates this protocol, not universal performance limits. Provider-verified observations, road context, and external validation are necessary before these estimates are extended to unobserved networks. The present study predicts detector passages, not passenger-car-only flows or the OD demand that generated them.

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Disclosure statement

The authors report no competing interests.

Author contributions

Nicola Bicocchi: conceptualisation and formal analysis. Marco Mamei: writing—original draft and writing—review and editing. All authors approved the final manuscript and agree to be accountable for all aspects of the work.

Declaration of generative AI use

Generative AI was used only for proofreading and preparing submission materials, not for data analysis or interpretation.

Data availability statement

The data and code supporting this study will be available after publication upon request to the corresponding author.

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Table 1. Car-rental filtering. Rules are applied in order; each excluded row is counted once. Outside-area rentals are not necessarily invalid.

	2023	2024
Initial records	224,436	229,509
Exact duplicates	0	0
Invalid departure/year	0	0
Endpoints outside study box	29,360	13,344
Invalid duration or below 30 s	0	0
Retained rentals	195,076	216,165

Table 2. Retained car rentals. [a]Median endpoint displacement, not route length. [b]Median rental duration. [c]Percentage returning within 100 m. [d]Percentage rented over 24 h.

Year	Period	n	km[a]	min[b]	Near[c]	Long[d]
2023	Day	163,425	2.46	20.0	4.2	2.0
2023	Night	31,651	2.71	16.2	2.5	0.8
2024	Day	181,131	2.61	24.9	4.0	1.6
2024	Night	35,034	2.79	19.2	3.8	0.7

Table 3. Jensen–Shannon distance between normalised day and night rental OD patterns. Values depend on grid resolution and sampling.

Cell (km)	2023	2024
0.5	0.577	0.581
1	0.339	0.330
2	0.195	0.194

Table 4. Detector audit. The first exclusions refer to device-day rows; completeness exclusions refer separately to candidate day and night periods. Final site eligibility is applied afterwards.

	2023	2024
Initial device-day rows	313,067	309,561
Not active loop type	1,089	1,098
Invalid ID/position/date	60	0
Invalid hourly counts	0	0
Conflicting device-day rows	0	0
Repeated device-day aliases	999	674
Clock-change device-days	1,711	1,683
Candidate device-days	309,208	306,106
Incomplete day periods	150,108	178,984
Incomplete night periods	97,530	115,604
Complete periods before site eligibility	370,778	317,624

Table 5. Observation coverage at eligible devices after quality and stability checks. Dates count days with at least one usable period; missing periods are not zero flows.

Year	Devices	Periods	Median dates	Range
2023	752	355,288	314	181–344
2024	744	299,544	303	1–339

Table 6. Buffered spatial test in 2023. WAPE in percent. Macro averages device errors; pooled weights by detections; cluster averages after pooling within spatial components. Very large macro values are retained, not typographical errors.

Model	Features	Macro	Pooled	Cluster
Boosting	No sharing	261,493.2	62.9	93.3
Boosting	Citywide sharing	180,078.5	62.1	91.7
Boosting	Local endpoints	180,078.5	62.1	91.7
Boosting	OD geometry	180,078.5	62.1	91.7
Poisson	No sharing	190,930.8	64.9	99.3
Poisson	Citywide sharing	191,236.3	64.6	99.1
Poisson	Local endpoints	195,640.6	64.6	99.1
Poisson	OD geometry	197,584.4	64.6	99.0
Scaling	250 m chord	31,263.5	95.4	123.9

Table 7. Excluded-device test in 2024 with the 2023 models unchanged. WAPE in percent, with the same definitions as Table 6. Test sharing and weather are contemporaneously observed.

Model	Features	Macro	Pooled	Cluster
Boosting	No sharing	88,061.8	61.6	91.5
Boosting	Citywide sharing	93,709.6	61.7	91.8
Boosting	Local endpoints	93,709.6	61.7	91.8
Boosting	OD geometry	93,709.6	61.7	91.8
Poisson	No sharing	71,483.6	63.6	99.7
Poisson	Citywide sharing	78,590.4	68.8	109.8
Poisson	Local endpoints	80,922.5	69.1	110.6
Poisson	OD geometry	80,900.3	69.0	110.5
Scaling	250 m chord	4,115.7	98.2	137.5

Table 8. Median device WAPE in percent for selected configurations. S_+ counts positive-total devices; S counts all observed test devices. Undefined all-zero device WAPEs are omitted, not replaced by zero.

Year	Model	Inputs	Median	S_+/S
2023	Boosting	No sharing	57.0	752/752
2023	Boosting	Geometry	56.0	752/752
2023	Poisson	No sharing	59.8	752/752
2023	Poisson	Geometry	59.3	752/752
2024	Boosting	No sharing	55.9	736/744
2024	Boosting	Geometry	55.8	736/744
2024	Poisson	No sharing	59.8	736/744
2024	Poisson	Geometry	60.3	736/744

Table 9. Paired sharing-minus-no-sharing differences in mean device WAPE, in percentage points. Negative values favour sharing. Percentile intervals resample spatial blocks. Extreme intervals reflect near-zero detector totals; they should not be read as precise citywide effect estimates.

Year	Model	Added input	Δ	95% interval
2023	Boosting	Geometry	-81414.7	[-279283.0, +78.1]
2023	Boosting	Citywide	-81414.7	[-279283.0, +78.1]
2023	Boosting	Endpoints	-81414.7	[-279283.0, +78.1]
2023	Poisson	Geometry	+6653.6	[-8.4, +22695.6]
2023	Poisson	Citywide	+305.5	[-11.1, +971.0]
2023	Poisson	Endpoints	+4709.8	[-5.5, +16103.3]
2024	Boosting	Geometry	+5647.8	[-3.4, +19185.2]
2024	Boosting	Citywide	+5647.8	[-3.4, +19185.2]
2024	Boosting	Endpoints	+5647.8	[-3.4, +19185.2]
2024	Poisson	Geometry	+9416.8	[+16.1, +31934.3]
2024	Poisson	Citywide	+7106.8	[+15.7, +24087.3]
2024	Poisson	Endpoints	+9438.9	[+16.7, +32007.6]

Table 10. Full-feature models using only rentals of at most six hours. WAPE in percent. Count samples and spatial splits are unchanged; the contextual baseline is the same as in the primary comparison.

Year	Model	Macro	Pooled
2023	Boosting	216856.8	62.2
2023	Poisson	197267.7	64.6
2024	Boosting	87645.9	61.9
2024	Poisson	81439.8	69.2

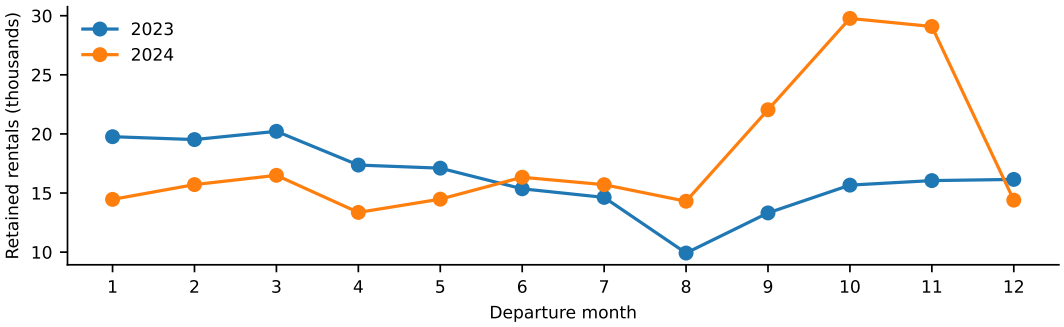

2023
2024
Retained rentals (thousands)
30
25
20
15
10
1
2
3
4
5
6
7
8
9
10
11
12
Departure month

Figure 1.

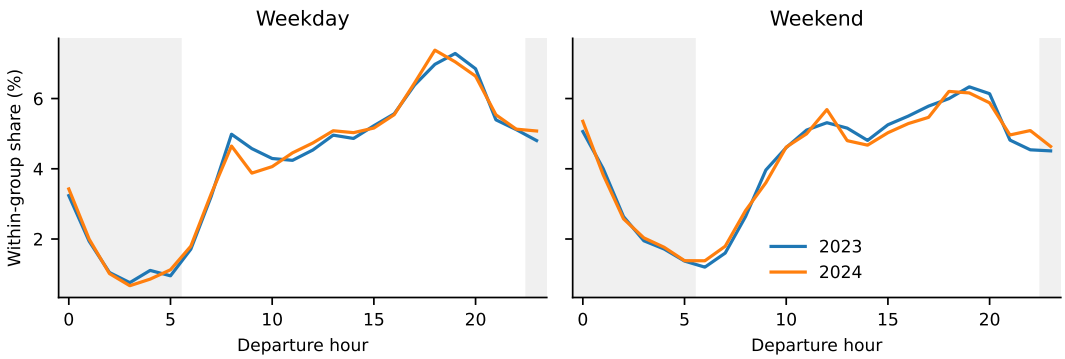

Weekday
Weekend
Within-group share (%)
6
4
2
0
5
10
15
20
Departure hour
2023
2024

Figure 2.

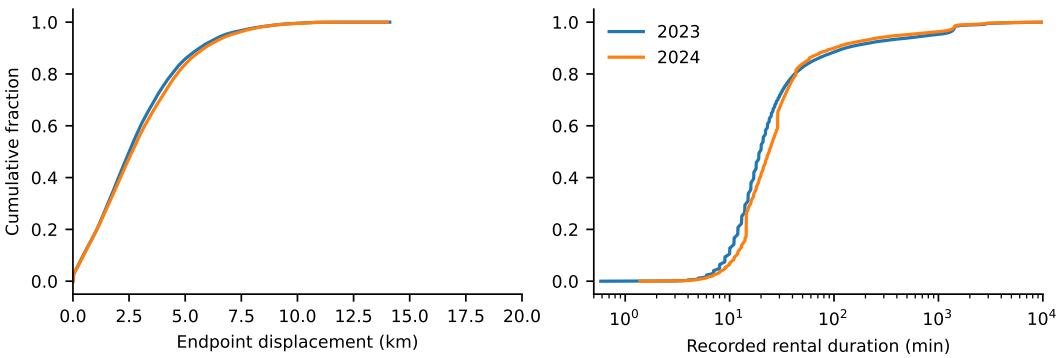

Cumulative fraction
1.0
0.8
0.6
0.4
0.2
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
Endpoint displacement (km)
2023
2024
10^0
10^1
10^2
10^3
10^4
Recorded rental duration (min)

Figure 3.

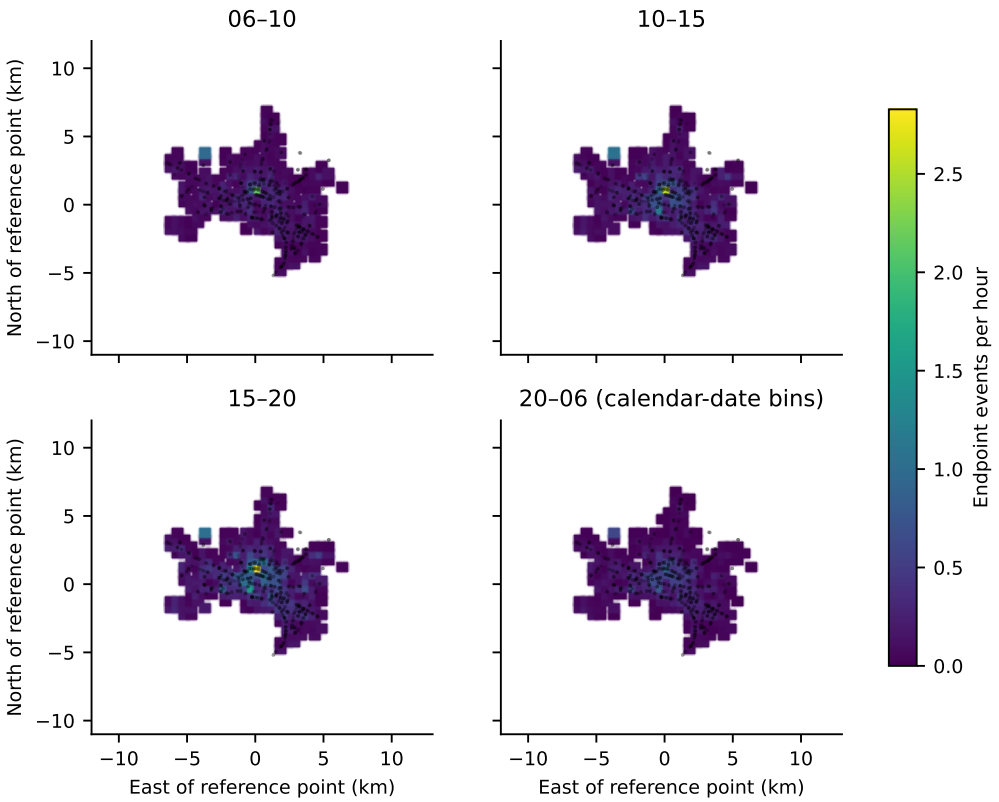

06–10
10–15
15–20
20–06 (calendar-date bins)
North of reference point (km)
East of reference point (km)
Endpoint events per hour
10
5
0
−5
−10
2.5
2.0
1.5
1.0
0.5
0.0

Figure 4.

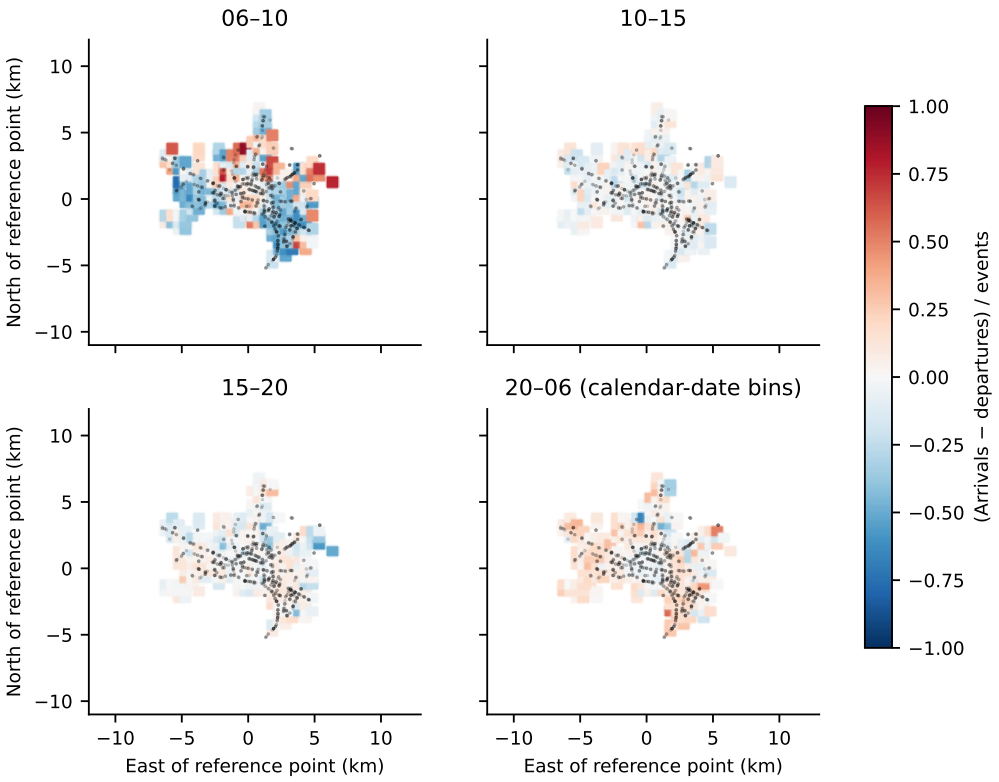

06–10
10–15
15–20
20–06 (calendar-date bins)
North of reference point (km)
East of reference point (km)
10
5
0
−5
−10
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
(Arrivals − departures) / events

Figure 5.

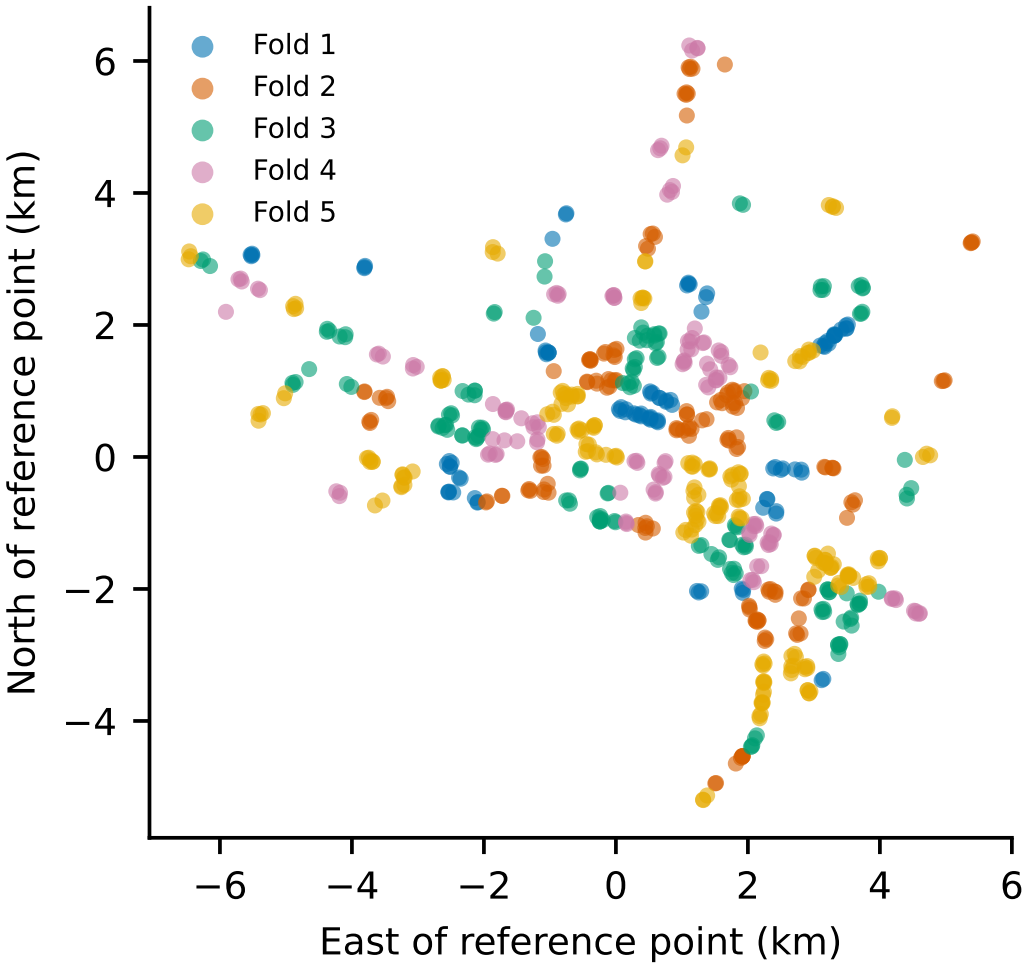

Fold 1
Fold 2
Fold 3
Fold 4
Fold 5
North of reference point (km)
East of reference point (km)
6
4
2
0
−2
−4
−6
−4
−2
0
2
4
6

Figure 6.

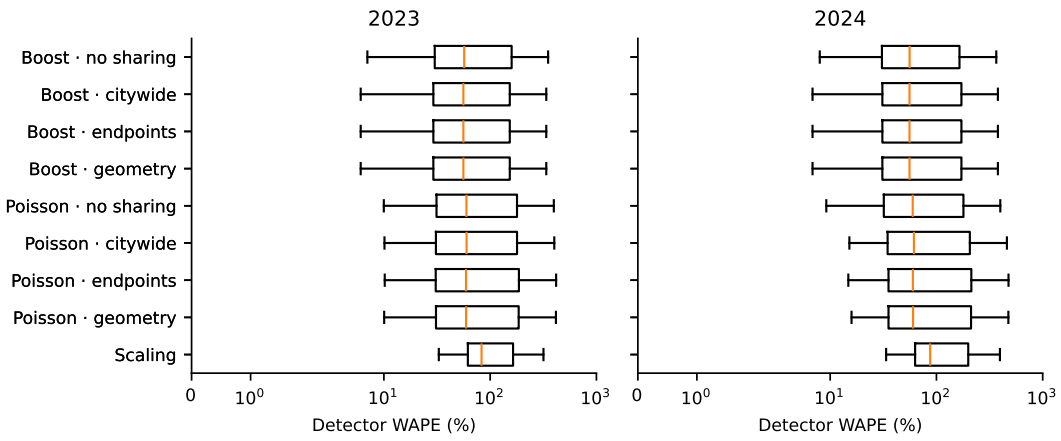

2023
2024
Boost · no sharing
Boost · citywide
Boost · endpoints
Boost · geometry
Poisson · no sharing
Poisson · citywide
Poisson · endpoints
Poisson · geometry
Scaling
0
10^0
10^1
10^2
10^3
Detector WAPE (%)

Figure 7.

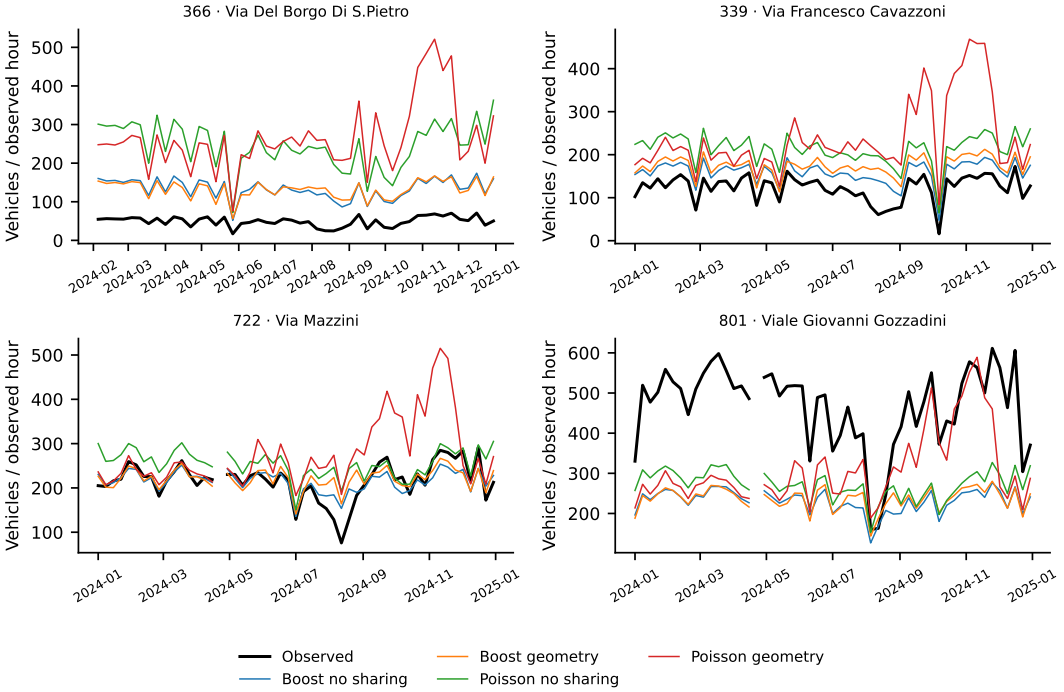

366 · Via Del Borgo Di S.Pietro
339 · Via Francesco Cavazzoni
722 · Via Mazzini
801 · Viale Giovanni Gozzadini
Vehicles / observed hour
Observed
Boost no sharing
Boost geometry
Poisson no sharing
Poisson geometry

Figure 8.

Figure captions

Figure 1. Monthly car-rental departures after the common endpoint and duration filters. Values describe the supplied sharing records, not total motor-vehicle demand.

Figure 2. Departure-hour profiles by year and day type. Each curve sums to 100% within its group. Shading marks the night bins, 00–06 and 23–24 on the same date.

Figure 3. Cumulative distributions of endpoint displacement and recorded rental duration. The displacement axis displays 0–20 km; the duration axis is logarithmic. Rental duration includes any time spent stopped or parked.

Figure 4. Car-sharing endpoint activity, pooled over 2023–2024: arrivals plus departures per included hour in 500 m cells. All panels share a scale. Black dots mark eligible detector positions; these are not observed rental crossings. Coordinates are kilometres from the central reference point. No road network is imputed.

Figure 5. Car-sharing arrival–departure balance in the same cells, dates and windows as Fig. 4. Red indicates more returns; blue indicates more collections. All panels share the $[-1, 1]$ scale and suppress cells below fifty events. Balance does not measure vehicles available for rental.

Figure 6. Eligible detector positions and their five spatial folds. Devices are grouped before block allocation. For each fold, training additionally excludes all devices within 250 m of any test device.

Figure 7. Distribution of device WAPE for the nine primary configurations. Boxes show quartiles and the median; whiskers extend to 1.5 times the interquartile range. Outlying points are not drawn and the displayed axis ends at 1,000%; the full, uncapped results are retained in Tables 6–7 and the numerical outputs. Devices with zero observed totals have undefined WAPE.

Figure 8. Observed and predicted 2024 weekly rates at four devices selected by 2023 flow quantiles. Each is excluded from its model's training fold, including the 250 m buffer. Counts and predictions are summed over available complete periods and divided by their exposure. Vertical scales differ; gaps indicate unavailable observations. These examples do not replace the all-device evaluation.